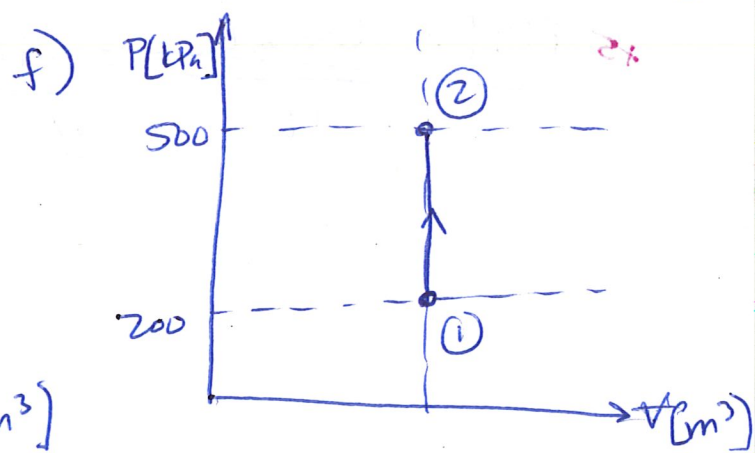
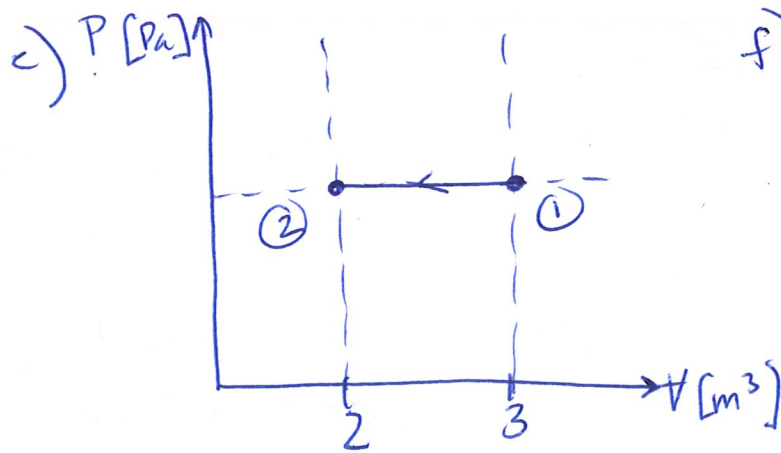
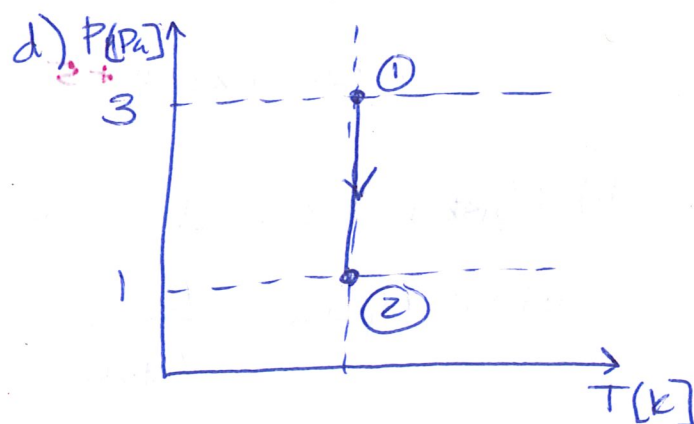
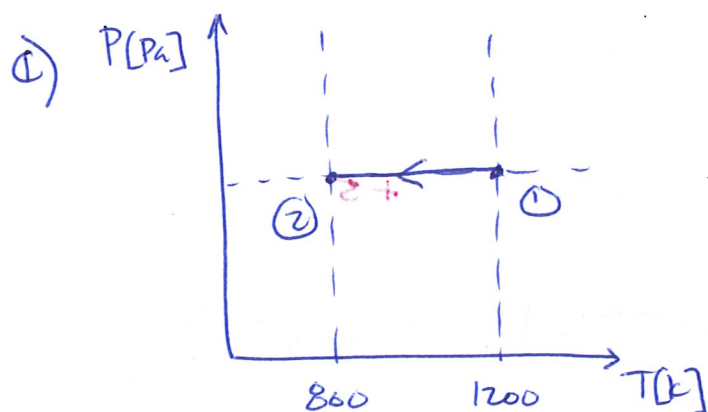
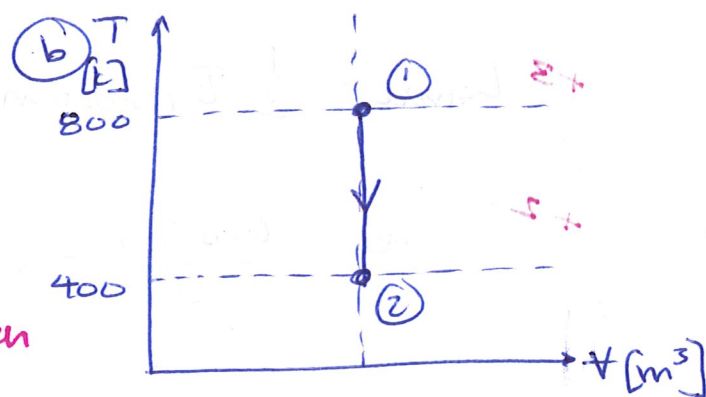
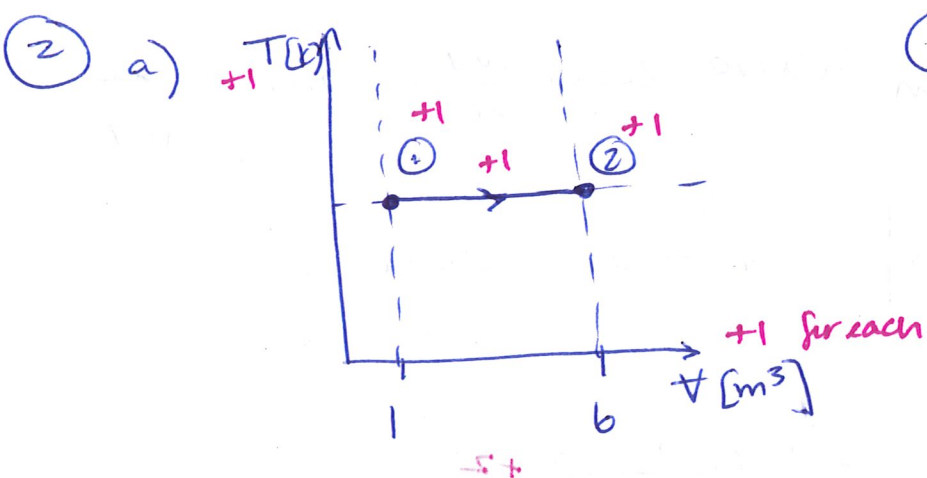


① Process - when a system undergoes a change from one state to another $\Rightarrow$ at the end, the system is at different at the end $+7$

Cycle - when a system undergoes a series of processes and the state of the system returns back to its original state. $+8$



③ a) Quasi-equilibrium: the process occurs slowly enough such that the system equilibrates much more quickly (on a shorter timescale) than the speed of the process (larger timescale). +5

+1 b) Bore $\rightarrow b = 89 \text{ mm}$

+1 Stroke $\rightarrow L = 100 \text{ mm}$

+1 Max speed $\rightarrow \text{RPM} = 6500 \text{ RPM}$

+3 $T_{\text{stroke}} = \frac{1}{2} T_{\text{revolution}}$ where $2000 \frac{\text{rev}}{\text{sec}} \Rightarrow 0.0005 \frac{\text{sec}}{\text{rev}}$

+2 $T_{\text{stroke}} = \frac{0.0005}{2} \Rightarrow T_{\text{stroke}} = 0.00025 \text{ sec}$

c) $T_{\text{stroke}} = \frac{1}{2} T_{\text{revolution}} = \frac{1}{2} \left(\frac{1}{\text{RPM}} \right)$ +2

$\rightarrow$ see excel +5

d) $U_{\text{piston}} = C_{\text{sound}}$ would be the limit +5

where $U_{\text{piston}} = \frac{L}{T_{\text{stroke}}} = \cancel{2 \text{ RPM}} \rightarrow T_{\text{max}} = \frac{L}{C_{\text{sound}}}$

+5 $\text{RPM} = \frac{C_{\text{sound}}}{2L} = \frac{340}{2(100 \times 10^{-3})} = \frac{0.1}{340} = 0.0003 \text{ s}$

$\text{RPM} \Rightarrow T_{\text{cycle}} = 2 \times T_{\text{max}} = 0.0006 \text{ s}$

$\Rightarrow 1666.7 \frac{\text{cycle}}{\text{s}}$

$\text{RPM} = 100,000$

④ Oxygen $\rightarrow$ $\text{O}-\text{O}$ $\Rightarrow$ modes of energy storage:
vibration, rotation, translation,
electronic +2

Neon $\rightarrow$ Ne $\Rightarrow$ modes of energy storage:
translation, electronic +2

$\Rightarrow$ because an oxygen molecule has more ways to store energy than a neon atom, then it has a higher heat capacity. +6

⑤ Based on completion +15

